MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 811A

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode

Mounting position: Vertical, base down or horizontal, pins 1 and 4 in vertical plane

Weight: 2.7 ounces nominal

Envelope: ST-19

Base: A4-10

Cap: C1-5

		Inches	MM
Height:	Maximum	6.469	164.31
Diameter:	*Maximum*	2.438	61.93

Base connections:

Pin No.	---	1	2	3	4	Cap
Element	---	f	nc	g	f	a

ABSOLUTE RATINGS: C Telegraphy

(D)

Parameter:	F1	Ef	Eb	Ec	Ib	Ic	Pp	Pi
Unit:	MHz	V	Vdc	Vdc	mAdc	mAdc	W	W
Maximum:	30	6.6	1,250	-200	175	50	45	175
Minimum:	---	6.0	---	---	---	---	---	---
TEST CONDITIONS:	---	6.3 Vdc	---	---	---	---	---	---

GENERAL:

Qualifications - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	3.75	4.25	Adc
1266	Total grid current	Eb = 2,500 Vdc; Ec/Ib = 18 mAdc (see note 1)	0.65	II	Ic	0	-5.0	μAdc
1256	Electrode current (anode)	Eb = 2,000 Vdc; Ec = -2 Vdc	0.65	II	Ib	16.0	36.0	mAdc
1231	Emission (1)	Ef = 4.5 Vdc; Eb = Ec = 100 Vdc (see note 1)	0.65	II	Is	80	---	mAdc
1236	Power oscillation (1)	Eb = 1,500 Vdc; Ib = 175 mAdc; Ic = 42 mAdc; F = 15 MHz; Rg = 3,500 ohms	0.65	II	Po	160	---	W
	Quality conformance inspection, part 2							
(D) 1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1036	Bump	Angle = 20°	---	---	---	---	---	---
1031	Low-frequency vibration	Eb = 250 Vdc; Ec = +7 Vdc; Rp = 2,000 ohms	---	---	Ep	---	500	mVac
1236	Power oscillation (2)	Eb = 1,100 Vdc; Ib = 140 mAdc; Ic = 50 mAdc (max); Ef = 6.3 Vac; F = 60 MHz; Rg = 3,500 ohms	---	---	Po	95	---	W
1236	Internal insulation		---	---	---	---	---	---
1231	Emission (2)	Power oscillation (1)	---	---	Ef	---	5.7	Vdc
1316	Amplification factor	Ec = 0, -2 Vdc; Eb/Ib = 20 mAdc; Mu = ΔEb/2	---	---	Mu	144	176	---
1331	Direct-interelectrode capacitance		--- --- ---	--- --- ---	Cgp Cin Cout	4.9 4.9 0.52	6.3 6.9 0.88	pF pF pF
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; Eb = 1,250 Vdc; Ib = 140 mAdc; Ic = 40 to 50 mAdc; Ef = 6.3 Vac; F = 30 MHz; Rg = 3,000 ohms; Po = 105 W (min); *t = 500 hours*	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3 -Continued							
---	Life-test end point	Emission (2)	---	---	Ef	---	6.0	Vdc

NOTE:

1. The tests listed below shall be the first tests performed after the holding period in the following order:

First: Emission (1)
Second: Total grid current

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MU
Navy - None
Air Force - 11, 70, 85

User activities:
Army - ME, AV, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

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